KARTHIKEYA M

CONTACT

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E-mail

Portfolio

Linked in

Leetcode

Github

Geeksforgeeks

Personal Blog

AREA OF INTEREST

- Massive MIMO
- Federated Learning
- Signal Processing
- Embedded System & Automation

TECHNICALSKILLS

- Signal Processing
- IoT Automation & Smart System
- PCB Design
- Front End Development
- Python
- c++

CAREER OBJECTIVE

To secure a challenging technical role in a dynamic environment where I can apply strong problem-solving and analytical skills, actively learn new technologies, continuously enhance my technical expertise, and contribute effectively to team objectives and organizational growth.

EDUCATION

- **B.E. Electronics and Communication Engineering (Hons)** — **2023-2027** **CGPA-7.9** (Till 5th semester)

  Velammal college of Engineering and Technology, Madurai.

- **HSC - State Board** — **2022-2023** **72%**

  Pasumalai Higher Secondary School, Madurai.

- **HINDI- (Praveen uttarardh)** — **2019-2022** **62.5%** **(First Class)**

  Dakshina Bharat Hindi Prachar Sabha,Madras.

- **SSLC - State Board** — **2020-2021** **(COVID)**

  Pasumalai Higher Secondary School, Madurai.

EXPERIENCE

- **Intern, CSUITE TECH LABS, Coimbatore.**

  Completed a two-week internship (June 16–28, 2025) learning the Cybersecurity Foundation & Protocols, Network Topologies, Network and Web Security, Cyber Awareness

- **Inplant Training, De Drone World Solution Private Limited, Coimbatore.**

  Involved in Inplant Training on (August-2024) & Learn about DGCA rules and regulations, Assembly of drone and learning the principles of aircraft landing and takeoff, along with exposure to CNS equipment relevant to Electronics and Communication Engineering.

SOFT SKILLS

- Regularity & Punctuality
- Adaptability & Curiosity
- Efficient Time Planner with Strong Self-Discipline
- Strong Interpersonal and Team Coordination Skills

CERTIFICATIONS

- Electronic Systems for Cancer diagnosis-NPTEL(Elite)
- Sensors & Actuators-NPTEL(Elite)
- Beginner Cyber Security – TATA Strive & Microsoft
- Industry 5.0

TOOLS

- CODESYS & Cisco
- Easy EDA
- MATLAB
- ESP32/Arduino
- HFSS/PSpice

ACHIEVEMENTS / AWARDS

- Won the Silver Medal in 50-meter Rifle Shooting at an NCC (National Cadet Corps) Camp, 2025.
- Secured the 3rd place in the Mathematics quiz competition(2024).
- 100% Attendance (2023-Till Date)

LANGUAGES KNOWN

- Tamil ★ ★ ★ ★
- English ★ ★ ★ ★
- Hindi ★ ★ ★

PROJECTS

- **Personal Portfolio Website (2025)**

  Develop and deployed a responsive professional portfolio to showcase technical skills and projects. Built using fundamental HTML, CSS, and JavaScript, serving as a dynamic, interactive digital resume.

- **Intelligent Integrated Wearable Safety System for Elderly Fall Prevention and Child Abuse (0CT-2025)**

  Built an IoT-based wearable safety system with fall detection and automatic alerting to improve safety for elderly and children.

- **Two way smart irrigation system (AUG-2025)**

  Designed an IoT-based smart irrigation system with automated water supply and drainage using soil-moisture sensing and remote monitoring to optimize water usage and prevent crop damage.

- **Automatic fire detection with IoT Alerts (APR-2024)**

  Designed an embedded fire detection and suppression system with automated water spraying and GSM-based emergency alerts.

KNOWLEDGE ENGAGEMENT

- Attended the CII (Confederation of Indian Industry) 8th Edition CONNECT Madurai 2025 – INVESTECH SOUTH, held on 28 November 2025 in Madurai.
- Completed a two-day technical workshop on "Empowering IoT," conducted by the ECE Department, Velammal College of Engineering and Technology, Madurai (August 2024).
- Participated in the DST-SERB sponsored Pre-Conference Workshop of VInCE'24 on "Cybersecurity Applications & NIR Spectrometers," held on 16 May 2024 at Velammal College of Engineering and Technology, Madurai.
- Successfully completed a two-day Generative AI workshop during Kurukshetra '24, held on 1–2 March 2024 at CEG, Anna University, Chennai, with hands-on exposure to LLMs and prompt engineering.

PATENT/RESEARCH

- Presented a research paper titled "An Overview: Clean Air, Healthy Care–AI-Driven Oxygen Purification in Hospitals" at the VAANI International Conference on Medical Devices, Diagnostics, Imaging & Digital Technologies (ICMDID-2024).
- Patent Filed (2024): Energy-harvesting and edge-enabled power system for sustainable WSNs in LPWAN deployments (Application No. 202441062415, filed 30 Aug 2024).